United States Patent	12408303
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lunsman; Harvey J. et al.

Elastomer embedded multipoint contact cooling

Abstract

Example implementations relate to a cooling assembly of a host circuit device, a circuit assembly including the host circuit device and a removable circuit device, and a method for thermal management of the removable circuit device removably connected to the host circuit device. The cooling assembly includes a cooling component and a thermal gap pad having an elastomer component movably connected to the cooling component and a plurality of beams embedded in the elastomer component. Each beam includes a first end portion, a second end portion, and a body portion extended between the first and second end portions. The first end portion of each of one or more beams is disposed in a first thermal contact with the cooling component and a second end portion of each of the one or more beams is disposed in a second thermal contact with a heat sink of the removable circuit device.

Inventors: **Lunsman; Harvey J. (Chippewa Falls, WI), Dean; Steven (Chippewa Falls, WI)**

Applicant: **HEWLETT PACKARD ENTERPRISE DEVELOPMENT LP** (Houston, TX)

Family ID: **89508696**

Assignee: **Hewlett Packard Enterprise Development LP (Spring, TX)**

Appl. No.: **17/815296**

Filed: **July 27, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240040748 A1	Feb. 01, 2024

Publication Classification

Int. Cl.: **H05K7/20** (20060101); **H05K1/02** (20060101)

U.S. Cl.:

CPC **H05K7/2039** (20130101); **H05K1/0203** (20130101);

Field of Classification Search

CPC: H05K (1/0203); H05K (1/181); H05K (7/2039); H05K (7/20472); H05K (7/20); H05K (7/14); H05K (7/20963); H05K (7/20445); H05K (7/20481); H05K (2201/10378); H05K (1/0204); H05K (7/20454); H05K (7/2049); H05K (7/20509); H05K (7/20463); H01L (23/3737); H01L (23/3736); H01L (23/36)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5159531	12/1991	Horvath	257/E23.09	H01L 23/3672
5365402	12/1993	Hatada	361/689	H01L 23/433
5371654	12/1993	Beaman	174/16.3	H01L 23/5384
5548090	12/1995	Harris	257/E23.09	H05K 7/20445
5695847	12/1996	Browne	428/389	H05K 1/0271
6311769	12/2000	Bonneville et al.	N/A	N/A
6854985	12/2004	Weiss	439/91	H01R 13/2414
7167379	12/2006	DiBene, II	257/691	G06F 1/182
8758892	12/2013	Bergin et al.	N/A	N/A
10651108	12/2019	Tang et al.	N/A	N/A
2003/0234451	12/2002	Razon	257/E23.068	H01L 24/85
2007/0204972	12/2006	Edward	361/705	F28F 3/022
2014/0146479	12/2013	Kilroy	361/717	H01L 23/433
2017/0164518	12/2016	Morgan	N/A	G02B 6/43
2022/0091642	12/2021	Lunsman	N/A	G06F 1/20
2022/0344239	12/2021	Lunsman et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108511407	12/2017	CN	N/A
2232969	12/2009	EP	N/A
2009/082732	12/2008	WO	N/A

Primary Examiner: Dravininkas; Adam B

Attorney, Agent or Firm: Hewlett Packard Enterprise Patent Department

Background/Summary

BACKGROUND

(1) An electronic device, such as a computer, a networking device, etc., may include a circuit assembly including a printed circuit board with a circuit module to provision the electronic device to communicate with an external circuit module. The circuit module may include a networking

switch, universal serial bus (USB) hub, or the like and the external circuit module may include a small form-factor pluggable (SFP) transceiver, a non-volatile memory express (NVMe) storage drive, or the like. The external circuit module, when connected to the circuit module, may produce waste heat during its operation. In order to minimize adverse effects of such waste heat on the external circuit module, the circuit assembly may include a cold plate to draw waste heat away from the external circuit module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various examples will be described below with reference to the following figures.

(2) FIG. **1**A illustrates an exploded perspective view of a thermal gap pad according to an example implementation of the present disclosure.

(3) FIG. **1**B illustrates an assembled perspective view of the thermal gap pad of FIG. **1**A according to an example implementation of the present disclosure.

(4) FIG. **1**C illustrates an exploded perspective view of a cooling component according to an example implementation of the present disclosure.

(5) FIG. **1**D illustrates an assembled perspective view of the cooling component of FIG. **1**C according to an example implementation of the present disclosure.

(6) FIG. **1**E illustrates a block diagram of a cooling assembly including the thermal gap pad of FIGS. **1**A-**1**B and the cooling component of FIGS. **1**C-**1**D according to an example implementation of the present disclosure.

(7) FIG. **2** illustrates a schematic diagram of a host circuit device having a host circuit board and a cooling assembly of FIG. **1**E according to an example implementation of the present disclosure.

(8) FIG. **3** illustrates a perspective view of a removable circuit device having a circuit board, a plurality of electronic components, and a heat sink according to an example implementation of the present disclosure.

(9) FIG. **4**A illustrates a perspective view of a portion of a circuit assembly of an electronic device having the removable circuit device of FIG. **3** removably connected to the host circuit device of FIG. **2** according to an example implementation of the present disclosure.

(10) FIG. **4**B illustrates a block diagram of a portion of the circuit assembly of FIG. **4**A having the cooling component of FIG. **1**C in thermal contact with the heat sink of FIG. **3** via the thermal gap pad of FIG. **1**B, and the circuit board of FIG. **3** communicatively connected to the host circuit board of FIG. **2** according to an example implementation of the present disclosure.

(11) FIG. **5** illustrates a flowchart depicting a method of thermal management of a removable circuit device according to an example implementation of the present disclosure.

DETAILED DESCRIPTION

(12) The following detailed description refers to the accompanying drawings. For purposes of explanation, certain examples are described with reference to the components illustrated in FIGS. **1**-**5**. The functionality of the illustrated components may overlap, however, and may be present in a fewer or greater number of elements and components. Moreover, the disclosed examples may be implemented in various environments and are not limited to the illustrated examples. Further, the sequence of operations described in connection with FIG. **5** is an example and is not intended to be limiting. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar parts. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only. While several examples are described in this document, modifications, adaptations, and other implementations are possible. Accordingly, the following detailed description does not limit the disclosed examples. Instead, the proper scope of the disclosed examples may be defined by the appended claims.

(13) As used herein, “host circuit device” may refer to a circuit module hosted on a circuit assembly of an electronic device. For example, the host circuit device may include a networking switch, a universal serial bus (USB) hub, or the like. As used herein, “circuit assembly” refers to an electronic circuit having a printed circuit board, the host circuit device, and a removable circuit device, where the removable circuit device and the host circuit device may function as a plug and a socket respectively of the electronic device. As used herein, “removable circuit device” may refer to an external circuit module which may be to be connected to the circuit assembly by way of plugging into the host circuit device to transmit data, receive data, store data, or process data. For example, the removable circuit device may be a pluggable transceiver device or a pluggable storage drive, or the like. Accordingly, the host circuit device may be a native device of the electronic device and the removable circuit device may be an ancillary device of the electronic device. As used herein, “electronic device” may refer to a computer (a server or a storage device), a networking device (a wireless access point or router), or the like. Further, as used herein, “plugging” of a removable circuit device into a host circuit device may refer to fitting the removable circuit device into the host circuit device by way of inserting or sliding the plug of the removable circuit device into the socket of the host circuit device. Further, as used herein, “thermal contact” may refer to a thermally coupling between the surfaces of two components to establish a thermally conductive pathway between the two components that allows heat to be conducted between the two components. Two objects may be considered to be thermally coupled if any of the following are true: (1) the two objects are in contact with one another (either direct contact, or contact via a thermal interface material), (2) the objects are both thermally coupled to a thermally conductive intermediary (e.g., a heat pipe, heat spreader, etc.) (or to a chain of thermally conductive intermediaries thermally coupled together), or (3) a heat transfer coefficient between the two objects is 5 W.Math.m.sup.−2.Math.K.sup.−1 or greater. For example, a first thermal contact may be formed by the contact of a first end portion of each of one or more beams of a thermal gap pad with a cooling component, and a second thermal contact may be formed by the contact of a second end portion of each of the one or more beams with a heat sink of the removable circuit device. As used herein, “cold plate” is sometimes used in the art with varying meanings, with some meanings being more generic and others being more specific. As used herein, “cold plate” refers specifically to a subset of thermal devices that are configured to receive waste heat from a component via conduction and to dissipate that waste heat into a flow of liquid coolant (e.g., water), in contrast to a “heat sink” which as used herein refers specifically to a subset of thermal devices that are configured to receive waste heat from a component via conduction and transfer that waste heat to the cold plate that is thermally coupled to the heat sink.

(14) An object, device, or assembly (which may include multiple distinct bodies that are thermally coupled, and may include multiple different materials), is “thermally conductive” between two thermal interfaces if any one of the following is true: (i) a heat transfer co-efficient between the thermal interfaces is 5 W.Math.m.sup.−2.Math.K.sup.−1 or greater at any temperature between 0° C. and 100° C., (ii) the object includes a material that has a thermal conductivity (often denoted k, λ, or κ) between the two interfaces is 1 W.Math.m.sup.−1.Math.K.sup.−1 or greater at any temperature between 0° C. and 100° C., or (iii) the object is a heat pipe, vapor chamber, body of copper, or body of aluminum. Examples of materials whose thermal conductivity greater than 1 W.Math.m.sup.−1.Math.K.sup.−1 between 0° C. and 100° C. include almost all metals and their alloys (e.g., copper, aluminum, gold, etc.), some plastics (e.g., TECACOMP® TC compounds, CoolPoly® D-series Thermally Conductive Plastics), and many other materials.

(15) An electronic device may include a circuit assembly to host a circuit module (e.g., a host circuit device). The host circuit device may provision the electronic device to communicate with an external circuit module (e.g., a removable circuit device). The removable circuit device may be removably connected to the host circuit device to perform various functions, such as transfer data, receive data, process data, store data, or the like. Typically, the removable circuit device converts

electrical signals into optical signals or vice versa, to perform the various functions discussed herein, and produces waste heat during its operation. If such waste heat is not removed from the removable circuit device, the waste heat may exceed thermal specifications of electronic components of the removable circuit device, thereby resulting in degraded performance, reliability, and life expectancy of the removable circuit device, and may also cause its failure.

(16) To overcome such issues related to waste heat in the removable circuit device, a heat spreader, i.e., a thermal transfer device that increases an area of thermal contact and/or distributes heat more evenly, may be thermally connected to the plurality of electronic components to transfer the waste heat from the removable circuit device. Further, cooling air may be blown over the heat sink to remove the waste heat from the heat spreader. However, when the removable circuit device is connected to the host circuit device, the heat spreader may not receive adequate supply of cooling air to remove the waste heat from the heat spreader, thereby degrading the performance of the removable circuit device.

(17) Hence, the host circuit device may include a cold plate to remove the waste heat from the heat spreader. For example, the cold plate may establish a thermal contact (or form a thermal interface) with the heat spreader when the removable circuit device is removably connected to the host circuit device to transfer the waste heat from the heat spreader to the cold plate, and thereby remove the waste heat from the heat spreader. However, it may be difficult to maintain the thermal contact between the cold plate and the heat spreader (e.g., between two interfacing surfaces), as the interfacing surfaces may not be flat and/or smooth. Also, the accumulation of debris and/or surface imperfections (e.g., scratches, dents, or the like) in either of the interfacing surfaces may compromise the heat transfer between the interfacing surfaces.

(18) To address the aforementioned issues with the interfacing surfaces, the host circuit device may further include a thermal interface material, such as thermally conductive grease. The thermally conductive grease may be disposed on one of the interfacing surfaces such that when the removable circuit device is removably connected to the host circuit device, the thermally conductive grease is interposed between the interfacing surfaces. Hence, the thermally conductive grease may maintain an optimal thermal contact between the cold plate and the heat spreader irrespective of variations in the interfacing surfaces. However, repetitive connection and disconnection between the removable circuit device and the host circuit device may cause the thermally conductive grease to be scraped off from one of the interfacing surfaces. Further, the thermally conductive grease may have to be replaced after every service event.

(19) In order to overcome the aforementioned issues with thermally conductive grease, the host circuit device may include a thermal interface material, such as thermally conductive multiple spring fingers. For example, one end of each of the multiple spring fingers may be coupled to the cold plate and other end of each of the multiple spring fingers may contact the heat spreader, when the removable circuit device is removably connected to the host circuit device. Each of the multiple spring fingers may exert a contact force (e.g., spring force) to establish the thermal contact with the heat spreader irrespective of variations in the surface of the heat spreader. However, the contact force exerted by each of the multiple spring fingers may oppose an insertion force applied by a user to connect the removable circuit device to the host circuit device or a removal force applied by a user to disconnect the removable circuit device from the host circuit device. In other words, each of the multiple spring fingers may have to exert an optimal contact force to establish a thermally conductive pathway between the heat spreader and the cold plate, while at the same time, allow the removable circuit device to be easily connected and disconnected by a user from the host circuit device. Moreover, a user may be required to apply repetitive force (e.g., multiple insertion forces and removal forces) to overcome the contact force between the removable circuit device and the host circuit device during repetitive connection and disconnection between the devices. Such repetitive application of force by the user could cause repetitive force related injuries to the user unless an optimal contact force between the spring fingers and the heat spreader is maintained

which is within acceptable safety limits. However, regulating the multiple spring fingers to maintain such an optimal contact force may be extremely difficult.

(20) A technical solution to the aforementioned problems may include providing a cooling assembly in a host circuit device of a circuit assembly to manage waste heat produced by a removable circuit device. In one or more examples, the cooling assembly includes a cooling component and a thermal gap pad including an elastomer component and a plurality of beams. The elastomer component is an elastic and thermally non-conductive component and each of the plurality of beams is a non-elastic and thermally conductive component. In some examples, the plurality of beams is embedded in the elastomer component such that a first end portion of each beam protrudes beyond a top surface of the elastomer component, a second end portion of each beam protrudes beyond a bottom surface of the elastomer component, and a body portion extending between the first and second end portions is retained in the elastomer component. The cooling component is rigidly connected to the cooling assembly and the elastomer component is movably connected to the cooling component. The thermal gap pad is disposed in the cooling assembly such that the first end portion of each of one or more beams of the plurality of beams is disposed in a first thermal contact with the cooling component and the second end portion of each of the one or more beams is disposed in a second thermal contact with a heat sink of the removable circuit device.

(21) In some examples, when the removable circuit device is removably connected to the host circuit device: i) the second end portion of each of the one or more beams rotates to contact the heat sink and displace the elastomer component upwards towards the cooling component, ii) the elastomer component moves each of the one or more beams upwards towards the cooling component, and iii) the first end portion of each of the one or more beams rotates to contact the cooling component and establish a thermally conductive pathway between the cooling component and the heat sink via each of the one or more beams. Thus, the second end portion of each of the one or more beams embedded in the elastomer component may establish a multi-point contact with the heat sink and the first end portion of each of the one or more beams embedded in the elastomer component may create another multi-point contact with the cooling component.

(22) The elastomer component may exert a contact force to establish and maintain the thermal contact between the heat sink and the cooling component via the one or more beams. In some examples, the contact force exerted by the elastomer component may be adjusted by changing the type of material used in the elastomer component. For example, a stiffer material used in the elastomer component or a thinner elastomer component may result in higher contact force. Further, the contact force exerted by the elastomer component may be tuned by changing physical properties or physical dimensions of the elastomer component. Thus, the elastomer component's ability to get displaced (e.g., bend or deform) upon connection of the removable circuit device may cause the one or more beams to change their physical geometry without impacting the contact force. Accordingly, a cross-sectional area of the one or more beams may be maximized to establish the thermally conductive pathway between the cooling component and the heat sink without the tradeoff of significantly higher contact force that the multiple spring fingers may require to establish the thermally conductive pathway.

(23) Further, the elastomer component may exert a multiplicity (array) of contact forces via the plurality of beams to create a substantially low insertion force for connecting the removable circuit device to the host circuit device and for disconnecting the removable circuit device from the host circuit device. At the same time, the elastomer component may provide the multipoint contact conduction cooling of the removable circuit device through the one or more beams for an effective thermal management of the removable circuit device. In some examples, the multiplicity of contact forces exerted by the elastomer component via the one or more beams is within acceptable safety limits to avoid repetitive force related injuries. For example, the contact force exerted by the elastomer component via each beam may be in a range from about 0.01 pound-force to 3.0 pound-

force. In some examples, the elastomer component may displace upwards towards the cooling component in a range from about 0.3 millimeter to 1.5 millimeter to allow easy plugging-in of the removable circuit device into the host circuit device. The elastomer component can maintain the multipoint contact with the heat sink via the plurality of beams, even though the cooling component and/or the heat sink has a non-smooth surface, a non-flat surface, surface imperfections, or debris, because the elastomer component may independently move each beam to establish the direct thermal interface with the cooling component and the heat sink. Accordingly, the elastomer component and the embedded plurality of beams provide multipoint conduction cooling to overcome the aforementioned problems related to multiple spring fingers.

(24) In one or more examples of the present disclosure, a cooling assembly for thermal management of a removable circuit device, and a host circuit device having such cooling assembly are disclosed. The cooling assembly includes a cooling component and a thermal gap pad. The thermal gap pad includes an elastomer component and a plurality of beams embedded in the elastomer component. The elastomer material is movably connected to the cooling component. Each of the plurality of beams includes a first end portion, a second end portion, and a body portion extended between the first and second end portions. The first end portion of each of the one or more beams of the plurality of beams is disposed in a first thermal contact with the cooling component and the second end portion of each of the one or more beams is disposed in a second thermal contact with a heat sink.

(25) FIG. **1**A depicts a perspective exploded view of a thermal gap pad **102** of a cooling assembly **100** (as shown in FIG. **1**E). FIG. **1**B depicts an assembled perspective view of the thermal gap pad **102**. In the description hereinafter, the Figures, FIGS. **1**A-**1**B are described concurrently for ease of illustration. The thermal gap pad **102** includes an elastomer component **106** and a plurality of beams **108** (as shown in FIG. **1**B). The thermal gap pad **102** may be used to establish a thermally conductive pathway between a cooling component **104** and a heat sink **310** (as shown in FIG. **3** and FIG. **4**B) for dissipation of waste heat from a plurality of electronic components **306** of the removable circuit device **300** (as shown in FIG. **3** and FIG. **4**B).

(26) In some examples, the elastomer component **106** is a rectangular-shaped elastic and thermally non-conductive component having a top surface **110** and a bottom surface **112**. The elastomer component **106** may be made of one of a silicone, a rubber, a foam, or a thermoplastic polyurethane material. The elastomer component **106** may displace (e.g., bend or deform) upon application of force (or load) on one of its surfaces and regain its original shape upon withdrawal of force. Additionally, the elastomer component **106** may not conduct (or transfer) the waste heat from the heat sink **310** to the cooling component **104**, even when the cooling component **104** and the heat sink **310** are disposed in thermal contact with each other via the elastomer component **106**. In some examples, the elastomer component **106** has a length “L.sub.1” that extends along a longitudinal direction **10**, a width “W.sub.1” that extends along a lateral direction **20**, and a height “H.sub.1” that extends along a vertical direction **30**. Further, the elastomer component **106** has a plurality of through-holes **114**. Each of the plurality of through-holes **114** extends between the top surface **110** and the bottom surface **112** of the elastomer component **106**. Further, each of the plurality of through-holes **114** are disposed spaced apart from each other along the longitudinal direction **30** and the lateral direction **20**. For example, the plurality of through-holes **114** are arranged along multiple columns **120** along the length “L.sub.1” to form an array of through-holes **115**. The array of through-holes **115** may include multiple set of through-holes, e.g., **115**A, **115**B . . . **115**N, which are arranged along multiple columns **120**. In some examples, a first set of through-holes **115**A among the array of through-holes **115** are arranged along a first column **120**A. Similarly, a second set of through-holes **115**B among the array of through-holes **115** are arranged along a second column **120**B among the plurality of columns **120**. In such examples, each hole in the first set of through-holes **115**A is positioned between two mutually adjacent through-holes in the second set of through-holes **115**B, and the first set of through-holes **115**A have an offset or staggered

arrangement with respect to the second set of through-holes **115**B. In some examples, each of the plurality of through-holes **114** is an angled hole having a rectangular-shaped cross sectional profile. Each hole of the plurality of through-holes **114** has a first width “W.sub.2”. In one or more examples, each through-hole among the plurality of through-holes **114** is inclined at a first angle “α.sub.1” relative to the bottom surface **112**. In some examples, the first angle “α.sub.1” may be in a range from about 30 degrees to about 60 degrees. The elastomer component **106** further includes a plurality of retention holes **116** (as clearly shown on FIG. **1**B). In some examples, each of the plurality of retention holes **116** has a fish-hook profile. The fish-hook profile of each retention hole includes a linearly extended through-hole portion **116**A and a curved hole portion **116**B connected to the linearly extended through-hole portion **116**A. The plurality of retention holes **116** are formed at the top surface **110** and positioned adjacent to peripheral corners **118** of the elastomer component **106**.

(27) In the example of FIG. **1**A, only one beam among the plurality of beams **108** is shown for ease of illustration. In one or more examples, each of the plurality of beams **108** has a slanting S-shaped profile, for example. In some examples, each of the plurality of beams **108** is made of one of a copper, an aluminum, an alloy material, or the like. Each of the plurality of beams **108** is a non-elastic and thermally conductive component. As used herein, “non-elastic component” may refer to a rigid component, which does not deform or bend upon application of force (or load). For example, the non-elastic component may only move (or rotate) upon application of force and may not deform or bend. Additionally, each of the plurality of beams **108** may conduct (or transfer) the waste heat from the heat sink **310** to the cooling component **104**, when the cooling component **104** and the heat sink **310** are disposed in thermal contact with each other via a corresponding beam **108**. In some examples, each of the plurality of beams **108** has a first end portion **122**, a second end portion **124**, and a body portion **126** extended between the first end portion **122** and the second end portion **124**. In some examples, the first end portion **122** may be referred to as a top end portion and the second end portion **124** may be referred to as a bottom end portion. In other words, the first end portion **122** and the second end portion **124** are opposite end portions of the corresponding beam **108**. In some examples, the first end portion **122** has a first curved surface **122**A and the second end portion **124** has a second curved surface **124**A that allows the first end portion **122** and the second end portion **124** of the corresponding beam **108** to rotate upon contacting a corresponding interfacing surface. The second end portion **124** of each of the plurality of beams **108** further includes an electrically insulating layer **128**, thereby preventing any electrical short-circuit. For example, the second end portion **124** may be coated with the electrically insulating layer **128**. In one or more examples, the electrically insulating layer **128** may be a thermally conductive layer. In some examples, the electrically insulating layer **128** may be made of one of a ceramic or a mica material. In some examples, each beam **108** has a second height “H.sub.2”, which is substantially greater than the first height “H.sub.1” of the elastomer component **106**. For example, the second height “H.sub.2” may be in a range from about 1.3 to 1.6 times greater than the first height “H.sub.1”. Further, each beam **108** has a second width “W.sub.3”, which is substantially equal to that of the first width “W.sub.2” of each through-hole **114** of the elastomer component **106**. In some examples, the body portion **126** of each beam **108** is an angled body portion. In one or more examples, the body portion **126** of each beam **108** is inclined at a second angle “β.sub.1” relative to the second end portion **124**. In some examples, the second angle “β.sub.1” may be in a range from about 30 degrees to about 60 degrees.

(28) Referring to FIG. **1**B, the elastomer component **106** is shown to be transparent for ease of illustration and such an illustration should not be construed as a limitation of the present disclosure. The plurality of beams **108** are embedded in the elastomer component **106**. For example, each beam **108** is disposed in a corresponding through-hole of the plurality of through-holes **114** to embed the corresponding beam **108** to the elastomer component **106**. Since the second width “W.sub.3” of each beam **108** is substantially equal to the first width “W.sub.2” of each through-

hole **114**, each beam **108** press-fits inside the corresponding through-hole **114** to embed in the elastomer component **106**. In some examples, the first end portion **122** of each beam **108** protrudes beyond the top surface **110** of the elastomer component **106**, the second end portion **124** of each beam **108** protrudes beyond the bottom surface **112** of the elastomer component **106**, and the body portion **126** of each beam **108** is positioned within the elastomer component **106** of the elastomer component **106**. Since the second height "H.sub.2" of each beam **108** is substantially greater than the first height "H.sub.1" of the elastomer component **106**, the first end portion **122** and the second end portion **124** protrude beyond the elastomer component **106**. In one or more examples, each of the plurality of beams **108** are discrete beams. As used herein, "discrete beams" may refer to separate beams, which are not directly connected or coupled to each other. The discrete beams among the plurality of beams **108** are arranged adjacent to each other along the lateral direction **20** and the longitudinal direction **10** of the elastomer component **106** to form an array of beams **109**. The array of beams **109** may include multiple set of beams, e.g., **109**A, **109**B, . . . **109**N, which are arranged along the multiple columns **120** (as shown in FIG. **1**B). In some examples, a first set of beams **109**A among the array of beams **109** are arranged along the first column **120**A. Similarly, a second set of beams **109**B among the array of beams **109** are arranged along the second column **120**B.

(29) In one or more examples, the second end portion **124** of each of the plurality of beams **108** may absorb the waste heat from the heat sink **310**, the body portion **126** may conduct the absorbed waste heat to the first end portion **122** of the corresponding beam **108**, and the first end portion **122** may transfer the waste heat to the cooling component **104**.

(30) FIG. **1**C depicts an exploded perspective view of a cooling component **104** of the cooling assembly **100**. FIG. **1**D depicts an assembled perspective view of the cooling component **104** of FIG. **1**C. In the description hereinafter, the Figures, FIGS. **1**C-**1**D are described concurrently for ease of illustration. In one or more examples, the cooling component **104** is a thermally conductive component made of one of a copper, an aluminum, or an alloy. The cooling component **104** may function as a cold plate. The cooling component **104** may include a body **130** and a housing **132**. The body **130** may include heat transfer features **134**, for example, fins or internal tubes. Further, the housing **132** may be attached atop the body **130** in a fluid tight manner or may be monolithic to the body **130** (e.g., the body **130** and the housing **132** are formed as a single unit). The housing **132** may include a coolant inlet **136** and a coolant outlet **138**. The body **130** may absorb the waste heat from the first end portion **122** of one or more beams among the plurality of beams **108**. The coolant may enter from the coolant inlet **136**, absorb heat from the body **130** via the heat transfer features **134**, and may exit from the coolant outlet **138**. The cooling assembly **100** may further include a thermally conductive grease **140**. In some examples, the thermally conductive grease **140** may be disposed on an outer surface of the body **130**. The waste heat may be conducted from the first end portion **122** of one or more beams **108** to the body **130** of the cooling component **104** via the thermally conductive grease **140**. The thermally conductive grease **140** may be attached to the cooling component **104** via a thermally conductive epoxy layer.

(31) FIG. **1**E depicts a block diagram of the cooling assembly **100** of a host circuit device **200**, having the thermal gap pad **102** of FIGS. **1**A-**1**B and the cooling component **104** of FIGS. **1**C-**1**D. In one or more examples, the cooling assembly **100** is used for thermal management of a removable circuit device **300** (as shown in FIG. **3**). For example, the cooling assembly **100** may be disposed within a housing of the host circuit device **200** (as shown in FIG. **2**) for thermal management of the removable circuit device **300**. Accordingly, when the removable circuit device **300** is removably connected to the host circuit device **200**, the cooling assembly **100** may establish a thermally conductive pathway between the heat sink **310** of the removable circuit device **300** and the cooling component **104** for conduction of waste heat from the plurality of electronic components **306** to the cooling component **104**.

(32) In some examples, the cooling component **104** may be rigidly coupled to a receptacle casing

208 of a host circuit device **200** (as shown in FIG. **2**). As discussed in FIGS. **1**C-**1**D, the cooling component **104** includes the body **130** and the housing **132**. The body **130** may include heat transfer features **134** (a shown in FIG. **1**D) and the housing **132** may be attached atop the body **130**. The housing **132** may include the coolant inlet **136** and the coolant outlet **138**. The coolant inlet **136** may be connected to an inlet manifold of a coolant distribution unit (CDU, not shown) and the coolant outlet **138** may be connected to an outlet manifold of the CDU.

(33) As discussed in FIGS. **1**A-**1**B, the thermal gap pad **102** includes the elastomer component **106** and the plurality of beams **108**. In one or more examples, the plurality of beams **108** are embedded in the elastomer component **106**. For example, the elastomer component **106** includes the plurality of through-holes **114**. In such examples, the plurality of beams **108** are disposed in the plurality of through-holes **114** to be embedded in the elastomer component **106**. The first end portion **122** of each beam **108** protrudes beyond the top surface **110** of the elastomer component **106**, the second end portion **124** protrudes beyond the bottom surface **112** of the elastomer component **106**, and the angled body portion **126** is disposed within the elastomer component **106**. Further, the elastomer component **106** having such plurality of embedded beams **108** is movably connected to the cooling component **104**. For example, the cooling assembly **100** includes a plurality of hooks **142** having a first end **142**A and a second end **142**B. The first end **142**A may have a flat profile and the second end **142**B may have a fish hook profile. In particular, the second end **142**B has a linearly extended portion **142**B.sub.1 and a curved portion **142**B.sub.2 connected to the linearly extended portion **142**B.sub.1. In such examples, the first end **142**A is rigidly connected to the cooling component **104** and the second end **142**B is disposed in a corresponding retention hole of the plurality of retention holes **116** formed in the elastomer component **106**. For example, the linearly extended portion **142**B.sub.1 of each hook **142** may be disposed in the linearly extended through-hole portion **116**A and the curved portion **142**B.sub.2 of each hook **142** may be disposed in the curved hole portion **116**B of a corresponding retention hole **116**. The linearly extended through-hole portion **116**A may allow the elastomer component **106** to reciprocate relative to the linearly extended portion **142**B.sub.1, and the curved portion **142**B.sub.2 may engage with the curved hole portion **116**B (as shown in FIG. **1**B) to movably connect the elastomer component **106** to the cooling component **104**. Therefore, such a connection of the elastomer component **106** to the cooling component **104** via the plurality of hooks **142** may allow the elastomer component **106** to have the movable connection relative to the cooling component **104**. In some examples, when the elastomer component **106** is movably connected to the cooling component **104**, the first end portion **122** of one or more beams of the plurality of beams **108** are disposed in a first thermal contact **144** with the cooling component **104** and the second end portion **124** of the one or more beams **108** are disposable in a second thermal contact **146** with the heat sink **310** of the removable circuit device **300**. In some examples, the first thermal contact **144** is formed between wet contact surfaces and the second thermal contact **146** is formed between dry contact surfaces. As used herein, “dry contact surfaces” may refer to thermally coupled surfaces of two objects, which does not contain any intervening elements there between to enable transfer of waste heat from one object to other object. As used herein, “wet contact surfaces” may refer to thermally coupled surfaces of two objects, which contains an intervening object such as a thermally conductive grease there between to enable transfer of waste heat from one object to the other object via the intervening object. In some examples, when the one or more beams **108** form the first and second thermal contacts with the cooling component **104** and the heat sink **310** respectively, a thermally conductive pathway **148** may be established between the heat sink **310** and the cooling component **104** via each of the one or more beams **108** to conduct the waste heat from the plurality of electronic components **306** to the cooling component **104**.

(34) FIG. **2** depicts a perspective view of a host circuit device **200**. In some examples, the host circuit device **200** is a networking switch, for example an Ethernet switch. In some other examples, the host circuit device **200** may be a universal serial bus (USB) hub, or the like without deviating

from the scope of the present disclosure. The host circuit device **200** may include a housing **202**, a host circuit board **204**, a plurality of sockets **206**, and a plurality of cooling assemblies **100**. It may be noted herein that the example of FIG. **2** shows only one socket **206** and one cooling assembly **100** for ease of illustration and such an illustration should not be construed as a limitation of the present disclosure. The host circuit device **200** may be an integral part of the circuit assembly **400** (as shown in FIG. **4**) or may be a modular component, which may be attached/coupled to the circuit assembly **400** of an electronic device such as a server system, a storage system, a networking system, or the like.

(35) The housing **202** may host the host circuit board **204** and include a plurality of receptacle casings **208**. In the illustrated example of FIG. **1**A, the host circuit device **204** has six receptacle casings **208**, which are disposed adjacent to each other. Further, the plurality of receptacle casings **208** are located at least partially over the host circuit board **204**. Each receptacle casing **208** includes a front end **210** having an opening **212** and a distal end **214**.

(36) The host circuit board **204** is disposed within the housing **202**. In some examples, the host circuit board **204** may include a front end section **220** and a rear end section **222**. In such examples, the rear end section **222** may be connected to a printed circuit board (e.g., mother board, not shown) of the circuit assembly **400** via a suitable interconnect mechanism, such as soldering.

(37) The plurality of sockets **206** are disposed spaced apart from each other and mounted on the front end section **220** of the host circuit board **204**. For example, each socket **206** is disposed proximate to the distal end **214** of the receptacle casing **208** such that an open end **216** of the corresponding socket **206** is disposed facing the opening **212** and the closed end **218** of the corresponding socket **206** is disposed on the front end section **220** of the host circuit board **204**. In some examples, each socket **206** may be a small form-factor pluggable (SFP) socket or an SFP port. Further, each of the plurality of sockets **206** facing the opening **212** may receive the removable circuit device **300** and establish a communication between the removable circuit device **300** and the circuit assembly **400** via the host circuit device **200**.

(38) In some examples, each cooling assembly **100** is disposed within a corresponding receptacle casing **208**, disposed adjacent to the host circuit board **204**, and located proximate to the front end **210** of the corresponding receptacle casing **208**. Further, each cooling assembly **100** is coupled to the corresponding receptacle casing **208**. For example, the cooling assembly **100** includes the cooling component **104**, which is rigidly connected to the corresponding receptacle casing **208** via a plurality of fasteners **224**. Each cooling assembly **100** further includes the thermal gap pad **102**. The thermal gap pad **102** includes a plurality of beams **108** and an elastomer component **106**. In one or more examples, the plurality of beams **108** are embedded in the elastomer component **106**. Further, the elastomer component **106** having the plurality of embedded beams **108** is movably connected to the cooling component **104** via a plurality of hooks **142**. In some examples, when the elastomer component **106** is movably connected to the cooling component **104**, the first end portion **122** of one or more beams of the plurality of beams **108** are disposed in the first thermal contact **144** (as shown in FIG. **1**E) with the cooling component **104** and the second end portion **124** of the one or more beams **108** are disposable in the second thermal contact **146** (as shown in FIG. **1**E) with the heat sink **310** of the removable circuit device **300**. In some examples, when the one or more beams **108** establishes the first and second thermal contacts with the cooling component **104** and the heat sink **310** respectively, a thermally conductive pathway **148** (as shown in FIG. **1**E) may be established between the heat sink **310** and the cooling component **104** via each of the one or more beams **108** to conduct the waste heat from the plurality of electronic components **306** to the cooling component **104**.

(39) FIG. **3** depicts a perspective view of a removable circuit device **300**. In some examples, the removable circuit device **300** is a pluggable electronic device, for example, a data communication device having a transceiver. In certain examples, the transceiver may be a small form-factor pluggable (SFP) transceiver coupled to an active optical cable (AOC, not shown) or a Quad small

form-factor pluggable (QSFP) transceiver coupled to the AOC, or the like. Other types of the removable circuit device **300**, such as the storage drive, for example, NVMe storage drive, or the like may be envisioned without deviating from the scope of the present disclosure.

(40) The removable circuit device **300** includes a casing **302**, a circuit board **304**, a plurality of electronic components **306**, a plug **308**, a heat sink **310**, and a handle **312**. The casing **302** may shield the circuit board **304** and the plurality of electronic components **306** from electromagnetic interference (EMI) and improve the reliability of the removable circuit device **300**. The casing **302** may have an opening **314** at a distal end **316** and the handle **312** connected at the front end **318**.

(41) In some examples, the circuit board **304** may include a semiconductor component. The circuit board **304** is disposed on a base on the casing **302**. The plurality of electronic components **306** is disposed on the circuit board **304**. In some examples, the plurality of electronic components **306** may include a processor, capacitors, resistors, or the like. The plug **308** is connected to a distal end (not labeled) of the circuit board **304** such that the plug **308** is disposed facing the opening **314**.

(42) In some examples, the heat sink **310** is located in the casing **302** and thermally coupled to the plurality of electronic components **306**. For example, the heat sink **310** is disposed on the plurality of electronic components **306** such that an inner surface **320** of the heat sink **310** is in thermal contact with the plurality of electronic components **306** and an outer surface **322** of the heat sink **310** is protruded beyond the removable circuit device **300**. The inner surface **320** of the heat sink **310** may absorb the waste heat from the plurality of electronic components **306** and transfer the absorbed waste heat to the outer surface **322**. In one or more examples, the heat sink **310** includes a thermally conductive material, for example, the copper material, the aluminum material, or the like. The handle **312** may be used to removably couple the removable circuit device **300** to the host circuit device **200**.

(43) FIG. **4**A depicts a perspective view of a portion of a circuit assembly **400** of an electronic device (not shown). FIG. **4**B depicts a block diagram of a portion of the circuit assembly **400** of FIG. **4**A having a cooling component **104**, a heat sink **310**, a circuit board **304**, and a host circuit board **204**. In the description hereinafter, the Figures, FIGS. **4**A-**4**B are described concurrently for ease of illustration. The circuit assembly **400** may be disposed within an enclosure **402** of the electronic device, such as a computer (a server or a storage device), a networking device (a wireless access point or a router), or the like. The circuit assembly **400** may include a host circuit device **200**, a printed circuit board (e.g., a mother board, not shown), and a removable circuit device **300**. In some examples, the host circuit device **200** and the printed circuit board may be disposed within the enclosure **402**. Further, the printed circuit board may be communicatively coupled to the host circuit device **200**, and the removable circuit device **300** may be removably connected to the host circuit device **200**.

(44) As discussed hereinabove in the example of FIG. **2**, the host circuit device **200** includes a host circuit board **204**, at least one socket **206**, and a cooling assembly **100**. The host circuit board **204** is disposed in a housing **202** (as shown in FIG. **2**) of the host circuit device **200**. The socket **206** is disposed within a receptacle casing **208** (as shown in FIG. **2**) of the housing **202**, and mounted on the host circuit board **204**. The cooling assembly **100** is disposed within and coupled to the receptacle casing **208**. The cooling assembly **100** includes a cooling component **104** rigidly connected to the receptacle casing **208**, and a thermal gap pad **102** movably connected to the cooling component **104**. The thermal gap pad **102** includes an elastomer component **106** and a plurality of beams **108** embedded in the elastomer component **106**.

(45) As discussed hereinabove in the example of FIG. **3**, the removable circuit device **300** includes a circuit board **304**, a plurality of electronic components **306**, a plug **308**, and a heat sink **310**. In the example implementation of the removable circuit device **300**, for illustration purposes, each of the plurality of electronic components **306** is shown as being a processor.

(46) The removable circuit device **300** may be removably connected to the host circuit device **200** via the opening **212** formed in the host circuit device **200**. In one or more examples, the removable

circuit device **300** is inserted (or plugged) inside the receptacle casing **208** via the opening **212** of the host circuit device **200** along the longitudinal direction **10**, to removably connect the removable circuit device **300** to the host circuit device **200**. In some examples, when the removable circuit device **300** is plugged (inserted) to the host circuit device **200**, the circuit board **304** is communicatively coupled to the host circuit board **204**. For example, the plug **308** of the circuit board **304** is inserted into socket **206** of the host circuit board **204** to communicatively couple the circuit board **304** to the host circuit board **204**. Further, when the removable circuit device **300** is plugged to the host circuit device **200**, the first end portion **122** of each of one or more beams of the plurality of beams **108** is disposed in a first thermal contact **144** with the cooling component **104**. In one or more examples, the first thermal contact **144** is formed between wet contact surfaces due to presence of the thermally conductive grease **140** between the first end portion **122** of each of the one or more beams **108** and the cooling component **104**. Further, the second end portion **124** of each of the one or more beams **108** is disposed in a second thermal contact **146** with the heat sink **310**. In one or more examples, the second thermal contact **146** is formed between the dry contact surfaces due to direct thermal contact between the second end portion **124** of each of the one or more beams **108** and the heat sink **310**. In one or more examples, when the first thermal contact **144** and the second thermal contact **146** are formed, a thermally conductive pathway **148** is established between the cooling component **104** and the heat sink **310** via each of the one or more beams **108**. For example, the thermally conductive pathway **148** is established between the heat sink **310** and the cooling component **104** via the first end portion **122**, the body portion **126** (as shown in FIG. **1**A), and the second end portion **124** of each of the one or more beams **108**. In one or more examples, the waste heat is conducted from the removable circuit device **300** to the host circuit device **200** via the thermally conductive pathway.

(47) In some examples, each beam is aligned at a first position **408** before the removable circuit device **300** is removably connected to the host circuit device **200**. When the removable circuit device **300** is connected to the host circuit device **200**, a contact force is exerted by the elastomer component **106** on the heat sink **310** via the plurality of beams **108**. In such examples, the second end portion **124** of each of the one or more beams **108** rotates along a counterclockwise direction **404** to contact the heat sink **310** and displace the elastomer component **106** upwards **406** towards the cooling component **104**. Accordingly, each of the one or more beams **108** move from the first position **408** to a second position **410** and displace the elastomer component **106** upwards towards the cooling component **104**. In one or more examples, the elastomer component **106** bends or deforms due to its displacement by the first end portion **122** of each of the one or more beams **108**.

(48) Further, when the elastomer component **106** is displaced upwards, it simultaneously moves each of the one or more beams **108** upwards towards the cooling component **104**. Since the body portion **126** of each of the one or more beams **108** is embedded in the elastomer component **106**, the displacement of the elastomer component **106** may result in displacement of the body portion **126** of each of one or more beams **108** via the elastomer component **106**.

(49) Later, the displacement of the body portion **126** of each of the one or more beams **108** may cause the first end portion **122** of each of the one or more beams **108** to rotate along a clockwise direction **412** to contact the cooling component **104**. Accordingly, the thermally conductive pathway **148** (as shown in FIG. **1**E) is established between the cooling component **104** and the heat sink **310** via each of the one or more beams **108**. In some examples, the first set of beams **109**A and the last set of beams **109**N among the array of beams **109** (only one beam in each set is shown for ease of illustration) may be rotated to an extent by the elastomer component **106** that they may not establish thermal contact with at least one of the heat sink **310** or the cooling component **104**. Since the first end portion **122** has a first curved surface **122**A and the second end portion **124** has the second curved surface **124**A, each of the one or more beams **108** are able to rotate easily to contact the cooling component **104** and the heat sink **310** respectively.

(50) In some examples, the contact force exerted by the elastomer component **106** may be regulated

based on at least one of a physical property (e.g., stiffness), a physical dimension (e.g., height, width, length) of the elastomer component **106**. Accordingly, the elastomer component's ability to get displaced (e.g., deform and bend) upon removable connection of the removable circuit device may cause the one or more beams to change their physical geometry without impacting the contact force. Accordingly, a cross-sectional area of the one or more beams **108** may be maximized to establish the thermally conductive pathway **148** and the heat conduction between the cooling component **104** and the heat sink **310**, without the tradeoff of significantly higher contact force that multiple spring fingers may require to establish a thermally conductive pathway and heat conduction.

(51) In one or more examples, the elastomer component **106** may exert an optimal contact force via each of the one or more beams **108** to allow easy plugging (e.g., inserting or sliding) of the removable circuit device **300** into the host circuit device **200**. For example, the elastomer component **106** may displace and bend marginally upwards towards the cooling component **104** when the removable circuit device **300** is plugged into the host circuit device **200**. However, the optimal contact force exerted by the elastomer component **106** via each of the one or more beams **108** may be sufficient to establish the thermally conductive pathway between the cooling component **104** and the heat sink **310**. In other words, the elastomer component **106** may provide a multiplicity (array) of the contact force via each of the one or more beams **108** to create a substantially low insertion force for plugging the removable circuit device **300** to the host circuit device **200**. In some examples, the multiplicity of the contact force is within acceptable safety limits to avoid repetitive force (e.g., insertion force or removal force) related injuries. In some examples, the amount of repetitive force that the removable circuit device **300** may require to removably connect to the host circuit device **200** or disconnect from the host circuit device **200** may be about 20 pound-force. The contact force exerted by the elastomer component **106** via each beam **108** may be in a range from about 0.01 pound-force to 3.0 pound-force. In some examples, the elastomer component **106** may displace upwards towards the cooling component **104** in a range from about 0.3 millimeter to 1.5 millimeter to allow easy plugging-in (insertion) of the removable circuit device **300** to the host circuit device **200**.

(52) In one or more examples, the elastomer component **106** may maintain the multipoint contact with the cooling component **104** and the heat sink **310** via one or more beams **108**, even though the interfacing surfaces have a non-smooth surface, a non-flat surface, surface imperfections, or debris, because the elastomer component **106** may exert optimal contact force via each beam **108** to establish the thermally conductive pathway **148** between the cooling component **104** and the heat sink **310**. It may be noted herein, that the interfacing surfaces may refer to i) an upper surface of the heat sink **310** and ii) a bottom surface of the cooling component **104**.

(53) During the operation, the plurality of electronic components **306** of the removable circuit device **300** may transmit, receive, process, or store data. Accordingly, the removable circuit device **300** may produce waste heat. In some examples, the removable circuit device **300** may produce the waste heat of about 20 joules per second. In such examples, the inner surface **320** of the heat sink **310**, which is thermally coupled to the plurality of electronic components **306** may transfer the waste heat from the plurality of electronic components **306** to the outer surface **322** of the heat sink **310**. Further, the second thermal contact **146** formed between the outer surface **322** of the heat sink **310** and the second end portion **124** of each of the one or more beams **108** (i.e., by way of dry contact surfaces) transfers the waste heat from the removable circuit device **300** to the thermal gap pad **102**. The waste heat is further transferred from the second end portion **124** to the first end portion **122** via the angled body portion **126**. Later, the first thermal contact **144** formed between the bottom surface of the cooling component **104** and the first end portion **122** of each of the one or more beams **108** (i.e., by way of a wet contact surfaces) transfers the waste heat from the thermal gap pad **102** to the cooling component **104**.

(54) The coolant may enter the cooling component **104** from the coolant inlet **136** (as shown in

FIGS. **1**C-**1**D), absorb waste heat from the body **130** (as shown in FIGS. **1**C-**1**D) via the heat transfer features **134** (as shown in FIGS. **1**C-**1**D), and may exit from the coolant outlet **138** (as shown in FIGS. **1**C-**1**D), thereby cooling the cooling component **104**. The heated liquid coolant may be pumped outside of the host circuit device **200** of the electronic device to exchange the heat with an external coolant (not shown) and regenerate the liquid coolant. Thus, in accordance to one or more examples of the present disclosure, the elastomer component **106** and the embedded plurality of beams **108** provide multipoint conduction cooling for an effective thermal management of the removable circuit device **300** and simultaneously overcome the problems related to multiple spring fingers.
(55) FIG. **5** is a flow diagram depicting a method **500** of a thermal management of a removable device. It should be noted herein that the method **500** is described in conjunction with FIGS. **1**A-**1**E, **2**, **3**, and **4**A-**4**B, for example.
(56) The method **500** starts at block **502** and continues to block **504**. At block **504**, the method **500** includes removably connecting a removable circuit device to a host circuit device, as described in FIGS. **4**A and **4**B. In some examples, the removable circuit device includes a circuit board, a plurality of electronic components disposed on the circuit board, and a heat sink thermally coupled to the plurality of electronic components. The host circuit device includes a host circuit board and a cooling assembly having a cooling component and a thermal gap pad including an elastomer component movably coupled to the cooling component and a plurality of beams embedded in the elastomer component. In some examples, removably connecting the removable circuit device to the host circuit device includes performing one of applying an insertion force to surpass a contact force applied by the elastomer component to establish the thermally conductive pathway, or applying a removal force to surpass the contact force to break the thermally conductive pathway.
(57) Further, the method **500** continues to block **506**. At block **506**, the method **500** includes the step of communicatively connecting the circuit board to the host circuit board, as described in FIGS. **4**A and **4**B. In some examples, the block **506** includes inserting a plug of the circuit board belonging to the removable circuit device into a socket of a host circuit board belonging to the host circuit device to communicatively couple the circuit board to the host circuit board. The method **500** continues to block **508**.
(58) At block **508**, the method **500** includes disposing a first end portion of each of one or more beams of the plurality of beams in a first thermal contact with the cooling component and a second end portion of each of the one or more beams in a second thermal contact with the heat sink to establish a thermally conductive pathway between the cooling component and the heat sink via each of the one or more beams, as described in FIGS. **4**A and **4**B. In one or more examples, the first thermal contact is formed between wet contact surfaces due to presence of the thermally conductive grease between the first end portion of each of the one or more beams and the cooling component. Further, the second thermal contact is formed between dry contact surfaces due to direct thermal contact between the second end portion of each of the one or more beams and the heat sink.
(59) In some examples, establishing the thermally conductive pathway includes rotating the second end portion of each of the one or more beams to contact the heat sink and displace the elastomer component upwards towards the cooling component and moving each of the one or more beams upwards towards the cooling component by the elastomer component. Additionally, establishing the thermally conductive pathway includes rotating the first end portion of each of the one or more beams to contact the cooling component. Accordingly, the thermally conductive pathway is established between the cooling component and the heat sink. The method **500** ends at block **512**.
(60) Various features as illustrated in the examples described herein may be implemented in a system, such as a host device and method for a thermal management of a removable device. In one or more examples, the elastomer component maintains an optimal contact force via one or more beams while plugging the removable circuit device to the host circuit device, which is within

acceptable safety limits to avoid repetitive force (e.g., insertion force or removal force) related injuries. Further, the elastomer component may be able to maintain the multipoint contact via the one or more beams with the removable circuit device and the cooling component, even though the surfaces of the cooling component and/or the heat sink has a non-smooth surface, a non-flat surface, surface imperfections, or debris.

(61) In the foregoing description, numerous details are set forth to provide an understanding of the subject matter disclosed herein. However, implementation may be practiced without some or all of these details. Other implementations may include modifications, combinations, and variations from the details discussed above. It is intended that the following claims cover such modifications and variations.

Claims

1. A cooling assembly of a host circuit device, comprising: a cooling component; and a thermal gap pad comprising an elastomer component movably connected to the cooling component and a plurality of beams embedded in the elastomer component, wherein a first end portion of each of one or more beams of the plurality of beams is disposed in a first thermal contact with the cooling component and a second end portion of each of the one or more beams is disposed in a second thermal contact with a heat sink of a removable circuit device, wherein the second end portion of each of the plurality of beams includes an electrically insulating layer, wherein the elastomer component is an elastic and thermally non-conductive component, and wherein each beam of the plurality of beams is a non-elastic and thermally conductive component.

2. The cooling assembly of claim 1, wherein the elastomer component comprises a plurality of through-holes extending between a top surface and a bottom surface of the elastomer component, and wherein the plurality of beams are disposed in the plurality of through-holes to be embedded in the elastomer component.

3. The cooling assembly of claim 2, wherein each through-hole of the plurality of through-holes is an angled hole, and wherein each beam of the plurality of beams comprises an angled body portion that extends between the first end portion and the second end portion of a corresponding beam.

4. The cooling assembly of claim 2, wherein the first end portion of each of the plurality of beams protrudes beyond the top surface of the elastomer component, and wherein the second end portion of each of the plurality of beams protrudes beyond the bottom surface of the elastomer component.

5. The cooling assembly of claim 1, wherein the elastomer component comprises at least one of silicone, rubber, foam, or thermoplastic polyurethane material, and wherein each beam of the plurality of beams comprises at least one of copper, aluminum, or an alloy.

6. The cooling assembly of claim 1, wherein the first end portion of each of the plurality of beams has a thermally conductive grease to establish the first thermal contact with the cooling component, and wherein the second end portion of each of the plurality of beams has a dry contact surface to establish the second thermal contact with the heat sink.

7. The cooling assembly of claim 1, wherein the plurality of beams are discrete beams, and wherein the discrete beams are arranged adjacent to each other along a lateral direction and a longitudinal direction of the elastomer component to form an array of beams.

8. The cooling assembly of claim 1, wherein a contact force is exerted by the elastomer component on the heat sink via the plurality of beams when the removable circuit device is removably connected to the host circuit device, and wherein the contact force is regulated based on at least one of a physical property or a physical dimension of the elastomer component.

9. The cooling assembly of claim 1, wherein the first end portion of each of the plurality of beams has a first curved surface and the second end portion of each of the plurality of beams has a second curved surface, and wherein, when the removable circuit device is removably connected to the host circuit device: i) the second end portion of each of one or more beams of the plurality of beams

rotates to contact the heat sink and displace the elastomer component upwards towards the cooling component, ii) the elastomer component moves each of the one or more beams upwards towards the cooling component, and iii) the first end portion of each of the one or more beams rotates to contact the cooling component and establish a thermally conductive pathway between the cooling component and the heat sink via each of the one or more beams.

10. A circuit assembly comprising: a removable circuit device comprising a circuit board, a plurality of electronic components disposed on the circuit board, and a heat sink thermally coupled to the plurality of electronic components; and a host circuit device comprising a host circuit board and a cooling assembly, wherein the removable circuit device is removably connected to the host circuit device, wherein the cooling assembly comprising: a cooling component; and a thermal gap pad comprising an elastomer component movably connected to the cooling component and a plurality of beams embedded in the elastomer component; wherein, when the removable circuit device is removably connected to the host circuit device: i) the host circuit board is communicatively coupled to the circuit board and ii) a first end portion of each of one or more beams of the plurality of beams is disposed in a first thermal contact with the cooling component and a second end portion of each of the one or more beams is disposed in a second thermal contact with the heat sink to establish a thermally conductive pathway between the cooling component and the heat sink via each of the one or more beams, wherein the elastomer component is an elastic and thermally non-conductive component, wherein each beam of the plurality of beams is a non-elastic and thermally conductive component, and wherein the second end portion of each of the plurality of beams includes an electrically insulating layer.

11. The circuit assembly of claim 10, wherein the elastomer component comprises a plurality of through-holes extending between a top surface and a bottom surface of the elastomer component, wherein the plurality of beams are disposed in the plurality of through-holes to be embedded in the elastomer component, wherein each through-hole of the plurality of through-holes is an angled hole, and wherein each beam of the plurality of beams comprises an angled body portion that extends between the first end portion and the second end portion of a corresponding beam.

12. The circuit assembly of claim 10, wherein the elastomer component comprises at least one of silicone, rubber, foam, or thermoplastic polyurethane material, and wherein each beam of the plurality of beams comprises at least one of copper, aluminum, or an alloy-material.

13. The circuit assembly of claim 10, wherein a contact force is exerted by the elastomer component on the heat sink via the plurality of beams when the removable circuit device is removably connected to the host circuit device, and wherein the contact force is regulated based on at least one of a physical property or a physical dimension of the elastomer component.

14. The circuit assembly of claim 10, wherein the first end portion of each of the plurality of beams has a first curved surface and the second end portion of each of the plurality of beams has a second curved surface, and wherein, when the removable circuit device is removably connected to the host circuit device: i) the second end portion of each of one or more beams of the plurality of beams rotates to contact the heat sink and displace the elastomer component upwards towards the cooling component, ii) the elastomer component moves each of the one or more beams upwards towards the cooling component, and iii) the first end portion of each of the one or more beams rotates to contact the cooling component and establish the thermally conductive pathway.

15. The circuit assembly of claim 10, wherein the removable circuit device is a pluggable electronic device comprising one of a small form-factor pluggable (SFP) transceiver or a non-volatile memory express (NVMe) storage drive.